MATP32 Project

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October 2026

1 Introduction

One of the first results which was shown in the course is the theorem about weak $(1,1)$ and strong (p,p) boundedness of the Hardy–Littlewood maximal function. In this project, we present a generalization of this theorem to sequences of locally integrable functions (or, from another perspective, ℓ^r–valued functions) which was first proven by C. Fefferman and E. M. Stein in 1971 [2]. Instead of the usual (∞,∞)–estimate however, the result we get concerns the so–called *exponential Orlicz space*; this can be understood as the maximal function of a bounded function not being too large (even if it still can be unbounded).

The main techniques which we use in the proof are the precisely the ones which have been used throughout the course, namely Marcinkiewicz interpolation, Calderón–Zygmund decomposition (and the associated decomposition of the function), and the usage of duality to obtain estimates.

2 Preliminaries

Recall first the definition of the Hardy–Littlewood maximal function

$$Mf(x) = \sup_{B \ni x \text{ ball}} \frac{1}{|B|} \int_B |f(y)| \mathrm{d}y.$$

For these proofs, it will instead be useful to consider the maximal function over cubes centered at x.

Definition 1. For $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, define

$$f^*(x) = \sup_{Q \text{ is a compact cube centered at } x} \frac{1}{|Q|} \int_Q |f(y)| \mathrm{d}y.$$

However, this is equivalent (up to constants) to the usual maximal function as we defined it in the course.

Lemma 2. *For $f \in L^1_{loc}(\mathbb{R}^d)$, there exist positive constants A, B depending only on the dimension, such that*

$$A \cdot Mf(x) \le f^*(x) \le B \cdot Mf(x) \quad \forall x \in \mathbb{R}^d$$

Proof. As we saw in the course, such equivalence constants exist between M and M_c, and also between M_c and $f \mapsto f^*$ (cf. problem 1 in [3]). Thus, the desired A and B exist by transitivity. □

The following inequality will also prove helpful.

Proposition 3 (Chebyshev's inequality)**.** *Let $(X, \mathscr{A}, \mu)$ be a measure space, let $f : X \to \mathbb{R}$ be measurable, and let $t > 0$, $0 < p < \infty$. Then,*

$$\mu(\{x \in X \mid |f(x)| > t\}) \leq t^{-p} \int_X |f|^p \mathrm{d}\mu.$$

Moreover, for a part of the theorem, we will need the following definition

Definition 4. Let $(X, \mathscr{A}, \mu)$ be a measure space, and let $\Phi : [0, \infty) \to [0, \infty)$ be a Young's function, i.e. Φ is continuous, increasing, convex, $\Phi(0) = 0$, and $\Phi(x) \to \infty$ as $x \to \infty$. Then, the *Orlicz norm* of a measurable $f : X \to \mathbb{R}$ is defined as

$$\|f\|_{\Phi(L)(X,\mu)} = \inf \left\{ \lambda > 0 \mid \int_X \Phi(|f|/\lambda) \mathrm{d}\mu \leq 1 \right\}.$$

The *Orlicz space* associated with Φ is defined as $\Phi(L)(X, \mu) = \{f \mid \|f\|_{\Phi(L)(X,\mu)} < \infty\}$.

It is not difficult to see that this is indeed a norm. Note also that the notation for the Orlicz space agrees with that of the L^p spaces, as they are precisely the Orlicz spaces for $\Phi(t) = t^p$.

Remark 5. The Orlicz space associated with $\Phi(t) = e^t - 1$ is usually denoted simply as $\exp(L)$, and not $\exp(L) - 1$.

A fact about Orlicz that we will use but not prove is that the dual space of $\exp L(\mathbb{R}^d)$ is $(L \log^+ L)(\mathbb{R}^d)$, i.e.

$$\Phi(t) = \begin{cases} 0 & \text{if } 0 \leq t \leq 1 \\ t \ln t & \text{otherwise} \end{cases},$$

in the sense that the functionals on $\exp L$ can be written as

$$f \mapsto \int_X f\phi \mathrm{d}\mu$$

for some $\phi \in L \log^+ L$. For the proof of this, see theorem 6.5 in chapter 4 of [1].

Finally, the proof of our main theorem will use a duality argument which depends on the following lemma.

Lemma 6. *Let $f, \varphi \in L^1_{loc}(\mathbb{R}^d)$ be positive. Then, for $r > 1$ there exists a constant $B_r > 0$ depending only on r and the dimension, such that*

$$\int_{\mathbb{R}^d} f^*(x)^r \varphi(x) \mathrm{d}x \leq B_r \int_{\mathbb{R}^d} |f(x)|^r \varphi^*(x) \mathrm{d}x.$$

Proof. For $k \in \mathbb{Z}$ let $\mathcal{D}_k$ be the family of dyadic cubes in $\mathbb{R}^d$ of side length 2^k, and let

$$f^{**}(x) = \sup_{k\in\mathbb{Z}, x\in Q\in\mathcal{D}_k} \frac{1}{|Q|}\int_Q |f(y)|\mathrm{d}y$$

be the dyadic maximal function. First, let us prove that

$$\int_{\mathbb{R}^d} f^{**}(x)^r\varphi(x)\mathrm{d}x \le B_r \int_{\mathbb{R}^d} |f(x)|^r\varphi^*(x)\mathrm{d}x$$

for some $B_r > 0$ and consider the operator $T(f) = f^{**}$. The statement we want to prove is thus the claim that T is a bounded operator from $L^r(\mathbb{R}^d, \varphi^*\mathrm{d}x)$ to $L^r(\mathbb{R}^d, \varphi\mathrm{d}x)$, with the bound also being independent of φ; this will be done using the Marcinkiewicz interpolation theorem as its bound only depends on r and the two edge–bound constants.

First, to show (∞,∞)–boundedness, note that for all x, $Tf(x) \le \|f\|_{L^\infty(\mathbb{R}^d,\lambda)}$ because the average of f over any set is always bounded by the essential supremum. Thus, $\|Tf\|_{L^\infty(\mathbb{R}^d,\varphi\mathrm{d}x)} \le \|f\|_{L^\infty(\mathbb{R}^d,\varphi^*\mathrm{d}x)}$ because the essential supremum with respect to these φ–weighted measures equals the essential supremum with respect to λ, this follows φ being positive, giving mutual absolute continuity of all three measures.

To show the weak $(1,1)$ bound, i.e. the existence of an A independent of φ such that

$$\int_{\{x\in\mathbb{R}^d | f^{**}(x)>\alpha\}} \varphi(y)\mathrm{d}y \le \frac{A}{\alpha}\int_{\mathbb{R}^d} f(x)\varphi^*(x)\mathrm{d}x,$$

we will use Calderón-Zygmund decomposition of f. This gives that $\{x \mid f^{**}(x) > \alpha\} = \cup_{Q\in\mathcal{Q}} Q$ where $\mathcal{Q}$ is a collection of pairwise disjoint dyadic cube with

$$\alpha < \frac{1}{|Q|}\int_Q |f(x)|\mathrm{d}x \le 2^d\alpha$$

for each $Q \in \mathcal{Q}$. Also, for such Q, we have that

$$\begin{aligned}\int_Q f(x)\varphi^*(x)\mathrm{d}x &\ge \int_Q f(x)\frac{2^{-d}}{|Q|}\int_Q \varphi(y)\mathrm{d}y\mathrm{d}x\\ &= \int_Q \varphi(y)\mathrm{d}y\int_Q f(x)\frac{2^{-d}}{|Q|}\mathrm{d}x\\ &\sim 2^{-d}\alpha\int_Q \varphi(y)\mathrm{d}y,\end{aligned}$$

where the last line means that it is bounded from both directions by independent constants (in particular 1 and 2^d). Then, summing over $Q \in \mathcal{Q}$ gives

$$\begin{aligned}\alpha\int_{\{x\in\mathbb{R}^d | f^{**}(x)>\alpha\}} \varphi(y)\mathrm{d}y &\le A\int_{\{x\in\mathbb{R}^d | f^{**}(x)>\alpha\}} f(x)\varphi^*(x)\mathrm{d}x\\ &\le A\|f\varphi^*\|_1,\end{aligned}$$

for some $A > 0$.

Now, define

$$f^*(x,N) = \sup_{Q \text{ cube centered at } x \text{ with } \ell(Q) < N} \frac{1}{|Q|} \int_Q |f(y)| \mathrm{d}y,$$

and note that it is monotone in N, with $\lim_{N\to\infty} f^*(x,N) = f^*(x)$. Thus, if we show that

$$\int_{\mathbb{R}^d} f^*(x,N)^r \varphi(x) \mathrm{d}x \leq B_r' \int_{\mathbb{R}^d} |f(x)|^r \varphi^*(x) \mathrm{d}x$$

for some B_r' independent of N, we will obtain the result of the lemma using the monotone convergence theorem. Moreover, by homogeneity (that is, a change of variables from x to Nx), it is enough to show the inequality for $N = 1$. Then, let τ_t be the family of translation operators, i.e. $\tau_t f(x) = f(x-t)$ and let $f_t^{**}(x) = (\tau_t f)^{**}(x+t)$. We claim that for some $c_d > 0$, $f^*(x,1) \leq c_d \int_{Q_0} f_t^{**}(x) \mathrm{d}t$, where Q_0 is the unit cube. To show the inequality, fix a cube Q centered at x of side length $0 < \ell < 1$. Moreover, let $k \in \mathbb{Z}$ and $s = 2^k$ with $\ell \leq s < 2\ell$, and note that $s \leq 1$. Then, let D_t be the unique dyadic cube of side length s containing $x + t$. By definition,

$$f_t^{**}(x) \geq \frac{1}{s^d} \int_{D_t - t} |f(y)| \mathrm{d}y \geq \frac{1}{s^d} \int_Q |f(y)| \mathrm{d}y \chi_{\{y,t | y+t \in D_t\}}(y,t) \mathrm{d}y.$$

Also, note that for each coordinate index $i \in \{1, \ldots, d\}$, we have that

$$\begin{aligned} |\{t_i \in [0,1) \mid x_i + t_i \text{ and } y_i + t_i \text{ share a dyadic interval of length } s\}| &= 1 - \frac{|x_i - y_i|}{s} \\ &\geq 1 - \frac{\ell/2}{s} \\ &\geq 1 - \frac{s/2}{s} = \frac{1}{2}, \end{aligned}$$

so

$$\int_{Q_0} \chi_{\{y+t \in D_t\}} \mathrm{d}t \geq 2^{-d}.$$

Thus, by Tonelli's theorem,

$$\begin{aligned} \int_{Q_0} f_t^{**}(x) \mathrm{d}t &\geq \int_{Q_0} \frac{1}{s^d} \int_Q |f(y)| \chi_{\{y+t \in D_t\}}(y,t) \mathrm{d}y \mathrm{d}t \\ &\geq \int_Q \frac{1}{s^d} |f(y)| \int_{Q_0} \chi_{\{y+t \in D_t\}}(y,t) \mathrm{d}t \mathrm{d}y \\ &\geq \frac{2^{-d}}{s^d} \int_Q |f(y)| \mathrm{d}y \\ &\geq \frac{2^{-d}}{(2\ell)^d} \int_Q |f(y)| \mathrm{d}y \\ &= 4^{-d} \frac{1}{|Q|} \int_Q |f(y)| \mathrm{d}y, \end{aligned}$$

and taking the supremum over Q gives the claimed inequality with $c_d = 4^d$.

From this, we can show the desired inequality as follows. By Jensen's inequality,

$$f^*(x,1)^r \leq c_d^r \left(\int_{Q_0} f_t^{**}(x) \mathrm{d}t \right)^r \leq c_d^r \int_{Q_0} f_t^{**}(x)^r \mathrm{d}t.$$

Therefore, using the first inequality we established, and the translation–invariance of the maximal function, we have that

$$\begin{aligned}
\int_{\mathbb{R}^d} f^*(x,1)^r \varphi(x) \mathrm{d}x &\leq \int_{\mathbb{R}^d} \left(c_d^r \int_{Q_0} f_t^{**}(x)^r \mathrm{d}t \right) \varphi(x) \mathrm{d}x \\
&= c_d^r \int_{Q_0} \int_{\mathbb{R}^d} f_t^{**}(x)^r \varphi(x) \mathrm{d}x \mathrm{d}t \\
&= c_d^r \int_{Q_0} \int_{\mathbb{R}^d} (\tau_t f)^{**}(x)^r \varphi(x-t) \mathrm{d}x \mathrm{d}t \\
&\leq c_d^r B_r \int_{Q_0} \int_{\mathbb{R}^d} (\tau_t f)(x)^r \varphi^*(x-t) \mathrm{d}x \mathrm{d}t \\
&= c_d^r B_r \int_{Q_0} \int_{\mathbb{R}^d} f(x)^r \varphi^*(x) \mathrm{d}x \mathrm{d}t
\end{aligned}$$

□

3 The theorem

Now, we can formulate the main theorem:

Theorem 7. *Let $(f_k)_{k\in\mathbb{N}}$ be a sequence of functions in $L^1_{loc}(\mathbb{R}^d)$. Then, for $1 < p, r < \infty$, we have that*

(i)

$$\left\| \left(\sum_{k=1}^{\infty} |f_k^*(x)|^r \right)^{1/r} \right\|_p \leq A_{r,p} \left\| \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right)^{1/r} \right\|_p$$

For some $A_{r,p}$ (which implicitly also depends on the dimension).

(ii)

$$\left| \left\{ x \in \mathbb{R}^d \mid \left(\sum_{k=1}^{\infty} |f_k^*(x)|^r \right)^{1/r} > \alpha \right\} \right| \leq \frac{A_r}{\alpha} \left\| \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right)^{1/r} \right\|_1$$

for $\alpha > 0$ and some A_r also depending on the dimension.

(iii)

$$\left(\sum_{k=1}^{\infty} |f_k(x)|^r \right)^{1/r} \in L^\infty(\mathbb{R}^d) \implies \left(\sum_{k=1}^{\infty} |f_k^*(x)|^r \right)^{1/r} \in \exp(L^r)$$

First, note that if one sets $f_k \equiv 0$ for $k > 1$, then Theorem 7(i) and (ii) reduce to the usual maximal function inequalities which were proven in the beginning of the course.

Now, let us prove the theorem.

Proof. We will consider the three cases $p = r$, $p < r$ and $p > r$ separately. Let us start with the first case. We have that, using the maximal inequality,

$$\begin{aligned}
\left\| \left(\sum_{k=1}^{\infty} |f_k^*(x)|^r \right)^{1/r} \right\|_r^r &= \int_{\mathbb{R}^d} \sum_{k=1}^{\infty} |f_k^*(x)|^r \mathrm{d}x \\
&= \sum_{k=1}^{\infty} \int_{\mathbb{R}^d} |f_k^*(x)|^r \mathrm{d}x \\
&= \sum_{k=1}^{\infty} \|f_k^*(x)\|_r^r \\
&\leq \sum_{k=1}^{\infty} A_r^r \|f_k(x)\|_r^r \\
&= A_r^r \left\| \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right)^{1/r} \right\|_r^r,
\end{aligned}$$

where A_r is the usual maximal function boundedness constant, and the exchange of integration and summation is justified by the integrand being positive (Monotone Convergence Theorem). This shows that (i) holds.

Now, we will show (ii). Let $\alpha > 0$ and apply the Calderón-Zygmund decomposition theorem on $F(x) := \left(\sum_{k=1}^{\infty} |f_k(x)|^r\right)^{1/r}$ (here, we assume that $F \in L^1(\mathbb{R}^d)$; if not, then the right-hand side of (ii) is infinite, and the inequality trivially holds) to obtain a countable (because there are only countably many dyadic cubes) collection of pairwise disjoint cubes $\mathcal{Q}$ with $F(x) \leq \alpha$ almost everywhere on $\mathbb{R}^d \setminus \cup\mathcal{Q}$ and

$$\alpha \leq \frac{1}{|Q|} \int_Q |F(x)| \mathrm{d}x \leq 2^d \alpha$$

for every $Q \in \mathcal{Q}$. By summing the first part of the latter inequality over $Q \in \mathcal{Q}$, we have that

$$\sum_{Q \in \mathcal{Q}} |Q| \leq \frac{1}{\alpha} \int_{\cup \mathcal{Q}} |F(x)| \mathrm{d}x \leq \frac{1}{\alpha} \int_{\mathbb{R}^d} |F(x)| \mathrm{d}x = \frac{1}{\alpha} \|F\|_1.$$

For $k \in \mathbb{N}$, let $f_k' = f_k \chi_{\mathbb{R}^d \setminus \cup \mathcal{Q}}$ and $f_k'' = f_k \chi_{\cup \mathcal{Q}}$. By the sub-additivity of $-^*$ and Minkowski's inequality on ℓ^r, we have that

$$\underbrace{\left(\sum_{k=1}^{\infty} |f_k^*(x)|^r \right)^{1/r}}_{:= G(x)} \leq \underbrace{\left(\sum_{k=1}^{\infty} |f_k'^*(x)|^r \right)^{1/r}}_{:= G'(x)} + \underbrace{\left(\sum_{k=1}^{\infty} |f_k''^*(x)|^r \right)^{1/r}}_{:= G''(x)}.$$

Now, because

$$|\{x \in \mathbb{R}^d \mid G(x) > 2\alpha\}| \leq |\{G' + G'' > 2\alpha\}| \leq |\{G' > \alpha\}| + |\{G'' > \alpha\}|,$$

it is enough to show that there are $A', A'' > 0$ with

$$|\{x \in \mathbb{R}^d \mid G'(x) > \alpha\}| \leq \frac{A'}{\alpha}\|F\|_1$$

and

$$|\{x \in \mathbb{R}^d \mid G''(x) > \alpha\}| \leq \frac{A''}{\alpha}\|F\|_1;$$

adding these together will give that

$$|\{x \in \mathbb{R}^d \mid G(x) > 2\alpha\}| \leq \frac{4\max(A', A'')}{2\alpha}\|F\|_1,$$

which is equivalent to (ii).

First, consider the former inequality, and let

$$F'(x) = \left(\sum_{k=1}^{\infty} |f_k'(x)|^r\right)^{1/r}.$$

We have that, by the properties of the Calderón-Zygmund decomposition,

$$\begin{aligned}
\|F'\|_r^r &= \int_{\mathbb{R}^d} F'(x)^r \mathrm{d}x \\
&= \int_{\mathbb{R}^d} F'(x)^{r-1} F'(x) \mathrm{d}x \\
&= \int_{\mathbb{R}^d \setminus \cup \mathcal{Q}} F(x)^{r-1} F(x) \mathrm{d}x \\
&\leq \alpha^{r-1} \int_{\mathbb{R}^d \setminus \cup \mathcal{Q}} F(x) \mathrm{d}x \\
&\leq \alpha^{r-1} \|F\|_1.
\end{aligned}$$

Applying the $p = r$ case of (i) which we have shown, we get that

$$\|G'\|_r^r \leq A_r^r \|F'\|_r^r \leq A_r^r \alpha^{r-1} \|F\|_1.$$

Thus, by Proposition 3, we have that

$$\begin{aligned}
|\{x \in \mathbb{R}^d \mid G'(x) > \alpha\}| &\leq \alpha^{-r} \|G'\|_r^r \\
&\leq \alpha^{-r} A_r^r \alpha^{r-1} \|F\|_1 \\
&= \frac{A_r^r}{\alpha} \|F\|_1,
\end{aligned}$$

which is precisely the desired inequality.

For the latter inequality, define

$$\widetilde{f}_k(x) = \begin{cases} \frac{1}{|Q|}\int_Q |f_k(y)|\mathrm{d}y & \text{if } x \in Q \text{ for some } Q \in \mathcal{Q}. \\ 0 & \text{otherwise} \end{cases}$$

and

$$\widetilde{F}(x) = \left(\sum_{k=1}^{\infty} |\widetilde{f}_k(x)|^r\right)^{1/r}.$$

Then, for $Q \in \mathcal{Q}$ and $x \in Q$, we have that

$$\begin{aligned} \widetilde{F}(x) = \left(\sum_{k=1}^{\infty} |\widetilde{f}_k(x)|^r\right)^{1/r} &= \left(\sum_{k=1}^{\infty} \left|\frac{1}{|Q|}\int_Q |f_k(y)|\mathrm{d}y\right|^r\right)^{1/r} \\ &= \frac{1}{|Q|}\left(\sum_{k=1}^{\infty} \left|\int_Q |f_k(y)|\mathrm{d}y\right|^r\right)^{1/r} \\ &\leq \frac{1}{|Q|}\int_Q \left(\sum_{k=1}^{\infty} |f_k(y)|^r\right)^{1/r} \mathrm{d}y \\ &= \frac{1}{|Q|}\int_Q F(y)\mathrm{d}y \\ &\leq 2^d\alpha, \end{aligned}$$

where the inequality exchanging the integral and the sum is an application of Minkowski's integral inequality on the σ–finite measure spaces $(\mathbb{R}^d, M_\lambda, \lambda)$ and $(\mathbb{N}, \mathcal{P}(\mathbb{N}), A \mapsto \#A)$. Now, because the support of $\widetilde{F}$ is in $\cup\mathcal{Q}$, we have, by the properties of the decomposition, that

$$\|\widetilde{F}\|_r^r \leq 2^{dr}\alpha^r|\cup\mathcal{Q}| = 2^{dr}\alpha^{r-1}\alpha\sum_{Q\in\mathcal{Q}}|Q| \leq 2^{dr}\alpha^{r-1}\|F\|_1.$$

Let $\gamma > 0$. Then, by Proposition 3,

$$|\{x \in \mathbb{R}^d \mid \widetilde{F}(x) > \gamma\alpha\}| \leq \gamma^{-r}\alpha^{-r}\|\widetilde{F}\|_r^r \leq \gamma^{-r}\frac{2^{dr}}{\alpha}\|F\|_1.$$

To show that the desired inequality for G'' follows from the established one for $\widetilde{F}$, we will apply a scaling trick. For any cube Q in $\mathbb{R}^d$, let $\widetilde{Q}$ be the cube with the same center as Q and $2d$ times the diameter (ensuring that for any two non–disjoint cubes, applying this operation to the larger one makes it contain the other), and let $\widetilde{\Omega} = \cup_{Q\in\mathcal{Q}}\widetilde{Q}$. Then,

$$|\widetilde{\Omega}| \leq \sum_{Q\in\mathcal{Q}}|\widetilde{Q}| = (2d)^d\sum_{Q\in\mathcal{Q}}|Q| \leq \frac{(2d)^d}{\alpha}\|F\|_1.$$

Now, we will show that for some fixed constant $K > 0$ and for $x \in \mathbb{R}^d \setminus \widetilde{\Omega}$,

$$f_k''^*(x) \leq K \widetilde{f}_k^*(x).$$

Let Q_0 be a cube (not necessarily one in $\mathcal{Q}$) centered at x, and let $J = \{Q \in \mathcal{Q} \mid Q_0 \cap Q \neq \varnothing\}$. Then,

$$\frac{1}{|Q_0|} \int_{Q_0} |f_k''(y)| \mathrm{d}y = \frac{1}{|Q_0|} \sum_{Q \in J} \int_{Q \cap Q_0} |f_k''(y)| \mathrm{d}y.$$

Moreover, for $Q \in J$ we have that because $Q \cap Q_0 \neq \varnothing$, either $Q \subseteq \widetilde{Q_0}$ or $Q_0 \subseteq \widetilde{Q}$. However, the latter case cannot happen because $x \in Q_0 \setminus \widetilde{\Omega} \subseteq Q_0 \setminus \widetilde{Q}$. Thus, we have that

$$\begin{aligned}
\frac{1}{|Q_0|} \sum_{Q \in J} \int_{Q \cap Q_0} |f_k''(y)| \mathrm{d}y &\leq \frac{1}{|Q_0|} \sum_{Q \in J} \int_{Q} |f_k''(y)| \mathrm{d}y \\
&= \frac{1}{|Q_0|} \sum_{Q \in J} \frac{|Q|}{|Q|} \int_{Q} |f_k''(y)| \mathrm{d}y \\
&= \frac{1}{|Q_0|} \sum_{Q \in J} |Q| \widetilde{f}_k(z_Q) \quad \text{for some } z_Q \in Q \\
&= \frac{1}{|Q_0|} \sum_{Q \in J} \int_{Q} \widetilde{f}_k(y) \mathrm{d}y \\
&\leq \frac{1}{|Q_0|} \int_{\widetilde{Q_0}} \widetilde{f}_k(y) \mathrm{d}y \\
&= \frac{|\widetilde{Q_0}|}{|Q_0|} \frac{1}{|\widetilde{Q_0}|} \int_{\widetilde{Q_0}} \widetilde{f}_k(y) \mathrm{d}y \\
&\leq (2d)^d \widetilde{f}_k^*(x),
\end{aligned}$$

which gives that $f_k''^*(x) \leq (2d)^d \widetilde{f}_k^*(x)$ and hence

$$G''(x) = \left(\sum_{k=1}^{\infty} |f_k''^*(x)|^r \right)^{1/r} \leq \left(\sum_{k=1}^{\infty} |(2d)^d \widetilde{f}_k^*(x)|^r \right)^{1/r} = (2d)^d \widetilde{F}(x)$$

for $x \in \mathbb{R}^d \setminus \widetilde{\Omega}$. Finally, to obtain the inequality for G'', we have that (using the $\widetilde{F}$–inequality with $\gamma = (2d)^{-d}$)

$$\begin{aligned}
|\{x \in \mathbb{R}^d \mid G''(x) > \alpha\}| &= |\{x \in \widetilde{\Omega} \mid G''(x) > \alpha\}| + |\{x \in \mathbb{R}^d \setminus \widetilde{\Omega} \mid G''(x) > \alpha\}| \\
&\leq \frac{(2d)^d}{\alpha} \|F\|_1 + |\{x \in \mathbb{R}^d \setminus \widetilde{\Omega} \mid (2d)^d \widetilde{F}(x) > \alpha\}| \\
&\leq \frac{(2d)^d}{\alpha} \|F\|_1 + (2d)^{dr} \frac{2^{dr}}{\alpha} \|F\|_1 \\
&= \frac{(2d)^d + (2d)^{dr} 2^{dr}}{\alpha} \|F\|_1.
\end{aligned}$$

This finishes the proof of (ii).

The case $p < r$ of (i) now follows from what we have shown and the Marcinkiewicz interpolation theorem. The weak–type $(1,1)$ estimate is precisely (ii), and the weak–type (r,r) one follows from (i) with $p = r$ and the fact that a strong estimate implies the corresponding weak one. It now only remains to show (i) with $p > r$ and (iii), which will be done using a duality trick.

Let $q > 1$ and let q' with $1/q + 1/q' = 1$. Then, for any positive $\varphi \in L^q$ with $\|\varphi\|_q = 1$, we have by Lemma 6 that

$$\begin{aligned}
\int_{\mathbb{R}^d} \sum_{k=1}^{\infty} f_k^*(x)^r \varphi(x) \mathrm{d}x &= \sum_{k=1}^{\infty} \int_{\mathbb{R}^d} f_k^*(x)^r \varphi(x) \mathrm{d}x \\
&\leq B_r \sum_{k=1}^{\infty} \int_{\mathbb{R}^d} |f_k(x)|^r \varphi^*(x) \mathrm{d}x \\
&= B_r \int_{\mathbb{R}^d} \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right) \varphi^*(x) \mathrm{d}x \\
&\leq B_r \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{q'} \|\varphi^*\|_q \quad \text{Hölder} \\
&\leq B_r \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{q'} C_{d,q} \|\varphi\|_q \quad \text{scalar H–L maximal ineq.} \\
&= B_r C_{d,q} \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{q'} .
\end{aligned}$$

Taking the supremum over all such φ gives that (by the duality between L^q and $L^{q'}$),

$$\left\| \sum_{k=1}^{\infty} f_k^*(x)^r \right\|_{q'} \leq B_r C_{d,q} \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{q'} .$$

Finally, set $p = rq'$. Because $q' > 1$, we have that $p > r$. Rewriting the inequality, we get that

$$\left\| \left(\sum_{k=1}^{\infty} f_k^*(x)^r \right)^{1/r} \right\|_p \leq B_r C_{d,q} \left\| \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right)^{1/r} \right\|_p ,$$

which is precisely the remaining case of (i).

Finally, for (iii), we will use a similar duality, but instead for $\exp L$ and $L \log^+ L$. If we assume that $(f_k)_{k=1}^{\infty}$ satisfies the assumptions of (iii), and let φ

range over the unit ball of the latter, we have that

$$\begin{aligned}
\left\| \sum_{k=1}^{\infty} f_k^*(x)^r \right\|_{\exp L} &= \sup_{\|\varphi\|_{L\log^+ L} \leq 1} \int_{\mathbb{R}^d} \left(\sum_{k=1}^{\infty} f_k^*(x)^r \right) \varphi(x) \mathrm{d}x \\
&\leq B_r \sup_{\varphi} \int_{\mathbb{R}^d} \left(\sum_{k=1}^{\infty} |f_k(x)|^r \right) \varphi^*(x) \mathrm{d}x \\
&\leq B_r \underbrace{\sup_{\varphi} \|\varphi^*\|_1}_{\text{bounded}} \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{\infty} \\
&\leq A_r \left\| \sum_{k=1}^{\infty} |f_k(x)|^r \right\|_{\infty},
\end{aligned}$$

which establishes (iii).

□

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